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Migrating data between byte-addressable and block-addressable storage devices in processor-based devices

Abstract

Migrating data between byte-addressable and block-addressable storage devices in processor-based devices is disclosed. In this regard, a processor of a processor-based device is communicatively coupled to both a byte-addressable storage device and a block-addressable storage device. The processor is configured to present the byte-addressable storage device and the block-addressable storage device as a single virtual storage device (i.e., as either a byte-addressable virtual device or a block-addressable virtual storage device). The processor is further configured to identify a low-activity region in the byte-addressable storage device, and to also identify a high-activity region in the block-addressable storage device. The processor then exchanges a first storage region corresponding to the low-activity region and comprising a memory address region of the byte-addressable storage device with a second storage region corresponding to the high-activity region and comprising a block region of the block-addressable storage device.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The technology of the disclosure relates generally to a storage system, and more particularly to a storage system having multiple tiers of storage media having different performance capabilities and different addressing mechanisms.

BACKGROUND

(2) Modern processor-based devices employ a variety of information storage systems to store, protect, and/or communicate data, and to provide user access to the data. Such information storage systems range from volatile memory devices (e.g., double data rate (DDR) random access memory (RAM) devices, synchronous dynamic RAM (SDRAM) memory devices, and the like, as non-limiting examples) to persistent storage devices (such as solid state drive (SSD) devices, NOT-AND (NAND) flash devices, and the like, as non-limiting examples). Memory devices provide faster access to stored data, but require power to maintain the stored data. Persistent storage devices, in contrast, provide relatively slower access to stored data, but are able to retain the stored data even when unpowered. Because conventional memory devices allow individual bytes of stored data to be directly accessed, they are referred to herein as “byte-addressable storage devices.” Persistent storage devices that allow direct access only to larger units (“blocks”) of data of a specified size (e.g., on the order of one (1) or more megabytes or larger) are referred to herein as “block-addressable storage devices.”

(3) To improve data access time using block-addressable storage devices, some conventional storage systems employ a tiering approach in which different types of block-addressable storage devices are classified into either a high-performance tier or a low-performance tier. The multiple tiers of block-addressable storage devices are presented to client devices as a single virtual block-addressable storage device. The storage systems monitor data access patterns and actively attempt to ensure that frequently accessed data is migrated into the high-performance tier and less frequently accessed data is migrated into the low-performance tier. Memory devices may be employed in such storage systems as caches for temporary storage of frequently accessed data, which may remain available only for a limited amount of time according to cache policies.

(4) However, it is desirable to further improve data access performance in processor-based devices based on current input/output (I/O) workload, to optimize media endurance, and to track lifetime expectancy of storage devices to ensure maximum uptime.

SUMMARY

(5) Exemplary embodiments disclosed herein include migrating data between byte-addressable and block-addressable storage devices in processor-based devices. In this regard, in one exemplary embodiment, a processor of a processor-based device is communicatively coupled to both a byte-addressable storage device (e.g., a memory device, as a non-limiting example) and a block-addressable storage device (e.g., a persistent storage device, as a non-limiting example). Both the byte-addressable storage device and the block-addressable storage device are used independently as direct access tiers behind a virtualization layer that presents both as a single virtual storage device

(i.e., as either a byte-addressable virtual device or a block-addressable virtual storage device). The processor is configured to identify a low-activity region in the byte-addressable storage device, and to also identify a high-activity region in the block-addressable storage device. The processor then exchanges a first storage region corresponding to the low-activity region and comprising a memory address region of the byte-addressable storage device with a second storage region corresponding to the high-activity region and comprising a block region of the block-addressable storage device.

(6) In some embodiments, exchanging the first storage region and the second storage region may comprise the processor copying the low-activity region to a first reserved section located on the block-addressable storage device using a byte-based read command to read from the byte-addressable storage device and a block-based write command to write to the block-addressable storage device. Some embodiments may provide that copying the low activity region may comprise converting from a byte format to a block format without losing data. The processor then updates a physical address of the low-activity region in a translation table. The processor next copies the high-activity region to a second reserved section located in the byte-addressable storage device using a block-based read command to read from the block-addressable storage device and a byte-based read command to write to the byte-addressable storage device. According to some embodiments, copying the high activity region may comprise converting from a block format to a byte format without losing data. The processor then updates a physical address of the high-activity region in the translation table.

(7) According to some embodiments, the processor may generate usage statistics for the byte-addressable storage device and the block-addressable storage device, and identify the low-activity region and the high-activity region based on the usage statistics. In some such embodiments, identifying the high-activity region based on the usage statistics may comprise comparing the usage statistics for the block-addressable storage device with a value of a promotion threshold.

(8) Some embodiments may provide that the processor receives a data access command to access a high-activity region in the block-addressable storage device. In embodiments in which the virtual storage device is presented as a byte-accessible virtual storage device, the data access command may comprise a byte-based storage command (i.e., a memory load command or a memory store command). Embodiments in which the virtual storage device is presented as a block-accessible virtual storage device may provide that the data access command comprises a block-based storage command. In such embodiments, the processor may be configured to identify the low-activity region and the high-activity region and exchange the first storage region with the second storage region responsive to receiving the data access command. According to some embodiments, the processor may be configured to identify the low-activity region and the high-activity region and exchange the first storage region with the second storage region at periodic time intervals (e.g., by executing a background process that wakes at regular time intervals).

(9) In another exemplary embodiment, a processor-based device is provided. The processor-based device comprises a virtualization circuit that is communicatively coupled to a byte-addressable storage device and a block-addressable storage device. The virtualization circuit is configured to present, to a processor of the processor-based device, the byte-addressable storage device and the block-addressable storage device as a single virtual storage device. The virtualization circuit is further configured to identify a low-activity region in the byte-addressable storage device. The virtualization circuit is also configured to identify a high-activity region in the block-addressable storage device. The virtualization circuit is additionally configured to exchange a first storage region corresponding to the low-activity region and comprising a memory address region of the byte-addressable storage device with a second storage region corresponding to the high-activity region and comprising a block region of the block-addressable storage device.

(10) In another exemplary embodiment, a method for migrating data between byte-addressable and block-addressable storage devices is provided. The method comprises presenting, by a processor of a processor-based device, a byte-addressable storage device and a block-addressable storage device

as a single virtual storage device. The method further comprises identifying, by the processor, a low-activity region in the byte-addressable storage device. The method also comprises identifying, by the processor, a high-activity region in the block-addressable storage device. The method additionally comprises exchanging, by the processor, a first storage region corresponding to the low-activity region and comprising a memory address region of the byte-addressable storage device with a second storage region corresponding to the high-activity region and comprising a block region of the block-addressable storage device.

(11) In another exemplary embodiment, a non-transitory computer-readable medium having stored thereon computer-executable instructions is provided. The computer-executable instructions, when executed by a processor, cause the processor to migrate data between byte-addressable and block-addressable storage devices by causing the processor to present a byte-addressable storage device and a block-addressable storage device as a single virtual storage device. The computer-executable instructions further cause the processor to identify a low-activity region in the byte-addressable storage device. The computer-executable instructions also cause the processor to identify a high-activity region in the block-addressable storage device. The computer-executable instructions additionally cause the processor to exchange a first storage region corresponding to the low-activity region and comprising a memory address region of the byte-addressable storage device with a second storage region corresponding to the high-activity region and comprising a block region of the block-addressable storage device.

(12) Those skilled in the art will appreciate the scope of the present disclosure and realize additional embodiments thereof after reading the following detailed description of the preferred embodiments in association with the accompanying drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

(1) The accompanying drawing figures incorporated in and forming a part of this specification illustrate several embodiments of the disclosure, and together with the description serve to explain the principles of the disclosure.

(2) FIG. 1 is a block diagram illustrating functional elements of a storage system for migrating data between byte-addressable and block-addressable storage devices, according to some embodiments;

(3) FIG. 2 is a block diagram illustrating exemplary virtual-to-physical address translations that may be performed by the storage system of FIG. 1, according to some embodiments;

(4) FIG. 3 is a block diagram illustrating an exchange of a high-activity region of the block-addressable storage device of FIGS. 1 and 2 with a low-activity region of the byte-addressable storage device of FIGS. 1 and 2, according to some embodiments;

(5) FIG. 4 is a block diagram illustrating an exemplary processor-based device including a virtualization circuit that is configured to migrate data between byte-addressable and block-addressable storage devices, according to some embodiments;

(6) FIG. 5 is a block diagram illustrating an exemplary processor-based device configured to execute a virtualization process to migrate data between byte-addressable and block-addressable storage devices, according to some embodiments;

(7) FIGS. 6A-6B provide a flowchart illustrating exemplary operations of the processor-based device of FIG. 4 or the processor-based device of FIG. 5 for migrating data between byte-addressable and block-addressable storage devices, according to some embodiments; and

(8) FIG. 7 is a block diagram of an exemplary processor-based device, such as the processor-based device of FIG. 4 or the processor-based device of FIG. 5, that is configured to migrate data between byte-addressable and block-addressable storage devices, according to some embodiments.

DETAILED DESCRIPTION

(9) Exemplary embodiments disclosed herein include migrating data between byte-addressable and block-addressable storage devices in processor-based devices. In this regard, in one exemplary embodiment, a processor of a processor-based device is communicatively coupled to both a byte-addressable storage device (e.g., a memory device, as a non-limiting example) and a block-addressable storage device (e.g., a persistent storage device, as a non-limiting example). Both the byte-addressable storage device and the block-addressable storage device are used independently as direct access tiers behind a virtualization layer that presents both as a single virtual storage device (i.e., as either a byte-addressable virtual device or a block-addressable virtual storage device). The processor is configured to identify a low-activity region in the byte-addressable storage device, and to also identify a high-activity region in the block-addressable storage device. The processor then exchanges a first storage region corresponding to the low-activity region and comprising a memory address region of the byte-addressable storage device with a second storage region corresponding to the high-activity region and comprising a block region of the block-addressable storage device.

(10) In some embodiments, exchanging the first storage region and the second storage region may comprise the processor copying the low-activity region to a first reserved section located in the block-addressable storage device using a byte-based read command to read from the byte-addressable storage device and a block-based write command to write to the block-addressable storage device. Some embodiments may provide that copying the low activity region may comprise converting from a byte format to a block format without losing data. The processor then updates a physical address of the low-activity region in a translation table. The processor next copies the high-activity region to a second reserved section located in the byte-addressable storage device using a block-based read command to read from the block-addressable storage device and a byte-based read command to write to the byte-addressable storage device. According to some embodiments, copying the high activity region may comprise converting from a block format to a byte format without losing data. The processor then updates a physical address of the high-activity region in the translation table.

(11) According to some embodiments, the processor may generate usage statistics for the byte-addressable storage device and the block-addressable storage device, and identify the low-activity region and the high-activity region based on the usage statistics. In some such embodiments, identifying the high-activity region based on the usage statistics may comprise comparing the usage statistics for the block-addressable storage device with a value of a promotion threshold.

(12) Some embodiments may provide that the processor receives a data access command to access a high-activity region in the block-addressable storage device. In embodiments in which the virtual storage device is presented as a byte-accessible virtual storage device, the data access command may comprise a byte-based storage command (i.e., a memory load command or a memory store command). Embodiments in which the virtual storage device is presented as a block-accessible virtual storage device may provide that the data access command comprises a block-based storage command. In such embodiments, the processor may be configured to identify the low-activity region and the high-activity region and exchange the first storage region with the second storage region responsive to receiving the data access command. According to some embodiments, the processor may be configured to identify the low-activity region and the high-activity region and exchange the first storage region with the second storage region at periodic time intervals (e.g., by executing a background process that wakes at regular time intervals).

(13) As used herein, the term “tier” refers to one or more storage devices that have a common performance characteristic. The term “performance characteristic” referred to herein is a measure of data handling capability that can include any device operation and interface timing. Examples of performance characteristics may include, as non-limiting examples, access time, data throughput, interface bandwidth, input/output operations per second (IOPS), and data priority. The term “storage device” used herein refers to an apparatus used for archiving and accessing data in a processor-based device environment. As non-limiting examples, a storage device may comprise a

magnetic disk drive, a solid state drive (SSD), an optical disk drive, a magnetic tape drive, or the like, without regard to the interface attachment used. The term “virtual storage device” is used herein to refer to the total addressable storage available to a processor-based device that is composed of one or more storage devices, each of which represents a portion of the total addressable storage. The virtual storage device is addressed by a virtual address that is translated to a physical address within the one or more storage devices.

(14) In this regard, FIG. 1 illustrates functional elements of a storage system **100** for migrating data between byte-addressable and block-addressable storage devices according to some embodiments. As seen in FIG. 1, the storage system **100** includes a storage processor **102** that comprises a processor interface **104** configured to implement a protocol for communicating with a process **106** executing on a processor of a processor-based device (not shown). The processor interface **104** may comprise, e.g., a parallel bus, a serial bus, an electronic connection, an optical connection, or a wireless connection, as non-limiting examples. The storage processor **102** is communicatively coupled to a byte-addressable storage device **108** via a byte-addressable storage interface **110**, and is likewise communicatively coupled to a block-addressable storage device **112** via a block-addressable storage interface **114**. The byte-addressable storage device **108** in some embodiments may comprise one or more memory devices, such as High Bandwidth Memory (HBM), Double Data Rate (DDR) Random-Access Memory (RAM) devices, Synchronous Dynamic RAM (SDRAM) memory devices, or Computer Express Link (CXL) external attached memory devices, as non-limiting examples. The block-addressable storage device **112** may comprise one or more NOT-AND (NAND) flash-based devices, standalone solid state disks (SSDs), CXL or fabric-attached external block-based storage devices, or very-high-performance serial-attached Small Computer System Interface (SCSI) (SAS) disk drives, as non-limiting examples.

(15) The storage processor **102** is configured to perform storage virtualization functions, including storage device tiering, virtual-to-physical device mapping, and other data management functions. In particular, the storage processor **102** provides an emulation and virtualization interface (EVI) **116** that controls transfers of bytes of data and configuration information to and from the byte-addressable storage device **108** via the byte-addressable storage interface **110**. The EVI **116** also controls transfers of blocks of data and configuration information to and from the block-addressable storage device **112** via the block-addressable storage interface **114**. The EVI **116** is configured to present the byte-addressable storage device **108** and the block-addressable storage device **112** to the process **106** as a single virtual storage device (i.e., as either a byte-addressable virtual storage device or a block-addressable virtual storage device). The virtual storage device in some embodiments may be configured as either a byte-addressable virtual storage device or a block-addressable virtual during setup of the virtualization functions, or may be automatically setup at power-up or reset of the storage processor **102** based on user preferences or policies. The storage processor **102** in some embodiments may be further configured to reconfigure the virtual storage device as a byte-addressable virtual storage device or a block-addressable virtual storage device without losing any data already stored on the byte-addressable storage device **108** and the block-addressable storage device **112**.

(16) Data written to a virtual address of the virtual storage device by the process **106** is initially stored in one of the byte-addressable storage device **108** or the block-addressable storage device **112** by the EVI **116**. As part of storing the data, the EVI **116** may convert from a block format to a byte format or vice versa by translating the virtual address into a physical address of a memory location of the byte-addressable storage device **108** or a physical address of a block region of the block-addressable storage device **112**. This may involve, e.g., translating a virtual address of a byte-based write command into a physical address of a block region of block-addressable storage device **112**, or translating a virtual address of a block-based write command into a physical address of a memory region of the byte-addressable storage device **108**. For example, if the virtual storage device is presented as a byte-addressable virtual storage device, the EVI **116** may receive a byte-

based write command from the process **106**. If the EVI **116** determines that the data should be written to the block-addressable storage device **112**, the EVI **116** may translate the virtual address specified by the byte-based write command into a physical address of a corresponding block region of the block-addressable storage device **112** by, e.g., utilizing read-modify-write (RMW) operations to modify the relevant byte portion of the block region. If the virtual storage device is presented as a block-addressable virtual storage device, the EVI **116** may receive a block-based write command from the process **106**. If the EVI **116** determines that the data should be written to the byte-addressable storage device **108**, the EVI **116** may translate the virtual address specified by the byte-based write command into a physical address of a corresponding memory region of the block-addressable storage device **112**. The EVI **116** employs a translation table **118** to store virtual-to-physical address translations for later access.

(17) The storage processor **102** further includes a data traffic statistics collection engine **120** that monitors data transmitted between the EVI **116** and the byte-addressable storage device **108** and the block-addressable storage device **112**. The data traffic statistics collection engine **120** collects data relating to traffic access patterns, such as most frequently read bytes or blocks of data, counts of read and write operations, streaming or random access characteristics, and the like, as non-limiting examples. The data traffic statistics collection engine **120** then generates and stores usage statistics **122** based on the collected data. The usage statistics **122** may be implemented, e.g., as a table of counters in RAM (not shown). This usage statistics **122** may be subsequently used for a variety of purposes, including determining low-activity regions and high-activity regions in each of the tiers based on their access frequency, endurance information based on the frequency of writes, or other user programmable criteria such as size of access requests.

(18) The storage processor **102** also provides a tiering decision engine **124** that uses usage statistics **122** to identify a low-activity region within the byte-addressable storage device **108** and a high-activity region within the block-addressable storage device **112**. For example, the tiering decision engine **124** may compare the usage statistics **122** (e.g., access frequency, as a non-limiting example) for a block region within the block-addressable storage device **112** with a promotion threshold (captioned as “PROM THRESH” in FIG. **1**) **126**. The promotion threshold **126** may comprise, e.g., an access frequency threshold, a data access latency threshold, an endurance threshold, and/or an access request size threshold, as non-limiting examples. If the tiering decision engine **124** determines that the usage statistics **122** for the block region within the block-addressable storage device **112** exceeds the promotion threshold **126**, the tiering decision engine **124** may identify the block region as the high-activity region. Some embodiments identify the low-activity region within the byte-addressable storage device **108** and the high-activity region within the block-addressable storage device **112** based on user policies and/or on physical media endurance and lifetime criteria, as non-limiting examples.

(19) To optimize the performance of the storage system **100**, the tiering decision engine **124** may determine that the low-activity region within the byte-addressable storage device **108** and the high-activity region within the block-addressable storage device **112** should be exchanged. The actual exchange of the data within the low-activity region for the data in the high-activity region is performed by a hybrid memory/block data migration and tiering engine **128** of the storage processor **102**. Exemplary operations that may be performed to accomplish the exchange of data are discussed in greater detail below with respect to FIG. **3**.

(20) It is to be understood that, while the storage system **100** of FIG. **1** is shown with two tiers of storage (i.e., the byte-addressable storage device **108** and the block-addressable storage device **112**), some embodiments may provide more tiers of storage. It is to be further understood that each tier of storage may comprise multiple storage devices.

(21) FIG. **2** illustrates an exemplary virtual-to-physical address translation that may be performed by the storage system **100** of FIG. **1**. Elements of FIG. **1** are referenced in describing FIG. **2** for the sake of clarity. In FIG. **2**, the byte-addressable storage device **108** and the block-addressable

storage device **112** of FIG. **1** are presented as a virtual storage device **200** (e.g., by the EVI **116** of FIG. **1**). The virtual storage device **200** is subdivided into multiple logical regions of equal size, such as logical region **202** and logical region **204**. Each of the logical regions of the virtual storage device **200** is mapped to a corresponding page of memory address regions of the byte-addressable storage device **108** or a corresponding page of block regions of the block-addressable storage device **112**. In the example of FIG. **2**, the logical region **202** of the virtual storage device **200** is mapped to a page **206** of memory address regions (such as memory address region **208**) of the byte-addressable storage device **108**, while the logical region **204** is mapped to a page **210** of block regions (e.g., block region **212**) of the block-addressable storage device **112**. The mappings may be stored in the translation table **118** of FIG. **1**.

(22) The byte-addressable storage device **108** and the block-addressable storage device **112** also include corresponding reserved sections **214** and **216**, respectively. The reserved sections **214** and **216** are set aside for use by the hybrid memory/block data migration and tiering engine **128** of the storage processor **102** of FIG. **1** when exchanging data between a low-activity region within the byte-addressable storage device **108** and a high-activity region within the block-addressable storage device **112**.

(23) In this regard, FIG. **3** illustrates an exchange **300** of a high-activity region **302** of the block-addressable storage device **112** of FIGS. **1** and **2** with a low-activity region **304** of the byte-addressable storage device **108** of FIGS. **1** and **2**, according to some embodiments. In the example of FIG. **3**, the exchange **300** is effectively a two-step transfer. First, the low-activity region **304** is copied into the reserved section **216** of the byte-addressable storage device **112**, as indicated by arrow **306**. Copying the low-activity region **304** may include, e.g., converting from a byte format to a block format without losing data. The physical address for the low-activity region **304** is then updated in the translation table **118** of FIG. **1** to reflect the new physical address for the low-activity region **304**. Next, the high-activity region **302** of the block-addressable storage device **112** is copied into the reserved section **214** of the byte-addressable storage device **108** (e.g., by converting from a block format to a byte format without losing data), as indicated by arrow **308**. The translation table **118** is then updated to reflect the new location of the high-activity region **302**. This exchange frees up the previously used low activity region **304** and high activity region **302** to be designated and used as the reserved section **214** and **216**, respectively, for a future data migration cycle. In some embodiments, the selection of the location of the reserved sections **214** and **216** within the byte-addressable storage device **108** and the block-addressable storage device **112**, respectively, may be based on user policies and/or on information provided by underlying physical media controllers where available (e.g., NAND spare locations or bad block locations, as non-limiting examples).

(24) According to some embodiments, additional reserved areas (not shown) on each tier may be provided for use during the transfer of data from one tier to another, and may include further operations to checkpoint the operations used during the transfer to enable recovery in the event of, e.g., power failure during the transfer. Some embodiments may provide that the exchange of the high-activity region **302** and the low-activity region **304** may be accomplished as a three-step transfer using a single reserved section by, e.g., copying the low-activity region **304** into the reserved section, copying the high-activity region **302** to the location in the byte-addressable storage device **108** that was occupied by the low-activity region **304**, and then copying the copy of the low-activity region in the reserved section into the block-addressable storage device **112**.

(25) According to some embodiments, the functionality of the storage processor **102** of FIG. **1** may be implemented as hardware. In this regard, FIG. **4** illustrates an exemplary processor-based device **400** that provides a processor **402** for migrating data between byte-addressable and block-addressable storage devices. The processor-based device **400** in some embodiments may comprise, e.g., an adapter card that is communicatively coupled to an application central processing unit (CPU) or a Compute Express Link (CXL) switch via a bus interface **404** to a computer bus **406**

(e.g., a CXL application computer bus). The processor **402** of FIG. **4** corresponds in functionality to the storage processor **102** of FIG. **1**, and may comprise, e.g., an enhanced CXL controller. The processor **402** includes a virtualization circuit **408** that implements the functionality of the EVI **116**, the data traffic statistics collection engine **120**, the tiering decision engine **124**, and the hybrid memory/block data migration and tiering engine **128** of FIG. **1**. The processor **402** is communicatively coupled to a memory device **410** that provides addressable memory used for data storage by the processor-based device **400**, and as such may comprise SDRAM or a non-volatile dual in-line memory module (NV-DIMM) device, as non-limiting examples. The memory device **410** may be used by the virtualization circuit **408** to store data such as the translation table **118** and/or the usage statistics **122** of FIG. **1**. In some embodiments, the translation table **118** and/or the usage statistics **122** may be also stored in other persistent memory or storage devices, such as in persistent reserved areas on the each of the byte-addressable storage device **412** and/or the block-addressable storage device **416** if present, for retrieval after a power cycle reset. The processor **402** is also communicatively coupled to a byte-addressable storage device **412** via a byte-addressable storage interface **414**, and likewise is communicatively coupled to a block-addressable storage device **416** via a block-addressable storage interface **418**.

(26) The processor-based device **400** of FIG. **4** and the constituent elements thereof may encompass any one of known digital logic elements, semiconductor circuits, processing cores, and/or memory structures, among other elements, or combinations thereof. Embodiments described herein are not restricted to any particular arrangement of elements, and the disclosed techniques may be easily extended to various structures and layouts on semiconductor sockets or packages. It is to be understood that some embodiments of the processor-based device **400** may include more or fewer elements than illustrated in FIG. **4**. For example, the processor **402** may further include one or more functional units, instruction caches, unified caches, memory controllers, interconnect buses, and/or additional memory devices, caches, and/or controller circuits, which are omitted from FIG. **4** for the sake of clarity.

(27) Some embodiments may provide that the functionality of the storage processor **102** of FIG. **1** is implemented in software (e.g., as a low-level kernel driver or application). Accordingly, FIG. **5** illustrates a processor-based device **500** that comprises a processor **502** corresponding in functionality to the storage processor **102** of FIG. **1**. The processor **502** executes a virtualization process **504** that implements the functionality of the EVI **116**, the data traffic statistics collection engine **120**, the tiering decision engine **124**, and the hybrid memory/block data migration and tiering engine **128** of FIG. **1**. The processor **502** of FIG. **5** is communicatively coupled to a memory device **506** (e.g., SDRAM or NV-DIMM, as non-limiting examples) that may be employed by the virtualization process **504** to store data such as the translation table **118** and/or the usage statistics **122** of FIG. **1**. In some embodiments, the translation table **118** and/or the usage statistics **122** may be also stored in other persistent memory or storage devices, such as in persistent reserved areas on the each of the byte-addressable storage device **508** and/or the block-addressable storage device **512** if present, for retrieval after a power cycle reset. The processor **502** is also communicatively coupled to a byte-addressable storage device **508** via a byte-addressable storage interface **510**, and is further communicatively coupled to a block-addressable storage device **512** via a block-addressable storage interface **514**.

(28) To illustrate exemplary operations of the processor-based device **400** of FIG. **4** or the processor-based device **500** of FIG. **5** for migrating data between byte-addressable and block-addressable storage devices according to some embodiments, FIGS. **6A-6B** provides a flowchart illustrating exemplary operations **600**. For the sake of clarity, elements of FIGS. **1-5** are referenced in describing FIGS. **6A-6B**. It is to be understood that some of the exemplary operations **600** illustrated in FIGS. **6A-6B** may be performed in an order other than that illustrated herein or may be omitted.

(29) The exemplary operations **600** begin in FIG. **6A** with a processor of a processor-based device

(e.g., the processor **402** of the processor-based device **400** of FIG. 4 or the processor **502** of the processor-based device **500** of FIG. 5) presenting a byte-addressable storage device (such as the byte-addressable storage device **108** of FIGS. 1-3) and a block-addressable storage device (such as the block-addressable storage device **112** of FIGS. 1-3) as a single virtual storage device (such as the virtual storage device **200** of FIG. 2), wherein the virtual storage device (**200**) is one of a block-addressable virtual storage device and a byte-addressable virtual storage device (block **602**). The virtual storage device **200** may be presented as a byte-addressable virtual device or may be presented as a block-addressable virtual device, based on configuration of the processor-based device **400**, **500** according to user preference or system policy. According to some embodiments, the processor **402**, **502** may generate usage statistics (e.g., the usage statistics **122** of FIG. 1) for the byte-addressable storage device **108** and the block-addressable storage device **112** (block **604**).

(30) Some embodiments may provide that the processor **402**, **502** receives a data access command to access a high-activity region (e.g., the high-activity region **302** of FIG. 3) in the block-addressable storage device **112** (block **606**). In embodiments in which the virtual storage device **200** is presented as a byte-accessible virtual storage device, the data access command may comprise a byte-based storage command (i.e., a memory load command or a memory store command). Embodiments in which the virtual storage device **200** is presented as a block-accessible virtual storage device may provide that the data access command comprises a block-based storage command.

(31) The processor **402**, **502** identifies a low-activity region (e.g., the low-activity region **304** of FIG. 3) in the byte-addressable storage device **108** (block **608**). Some embodiments may provide that the operations of block **608** for identifying the low-activity region **304** are based on the usage statistics **122** (block **610**). The processor **402**, **502** also identifies the high-activity region **302** in the block-addressable storage device **112** (block **612**). According to some embodiments, the operations of block **612** for identifying the high-activity region **302** may be based on the usage statistics **122** (block **614**). In some embodiments, the operations of block **614** for identifying the high-activity region **302** based on the usage statistics **122** may comprise comparing the usage statistics **122** for the block-addressable storage device **112** with a value of a promotion threshold (such as the promotion threshold **126** of FIG. 1) (block **616**). The exemplary operations **600** then continue at block **618** of FIG. 6B. The exemplary operations **600** shown in FIG. 6B may be performed immediately subsequent to the operations of block **616**, or may be performed later by a background function or process.

(32) Turning now to FIG. 6B, the processor **402**, **502** exchanges a first storage region corresponding to the low-activity region **304** and comprising a memory address region (e.g., the memory address region **208** of FIG. 2) of the byte-addressable storage device **108** with a second storage region corresponding to the high-activity region **302** and comprising a block region (such as the block region **212** of FIG. 2) of the block-addressable storage device **112** (block **618**). Some embodiments may provide that the operations of block **618** for exchanging the first storage region and the second storage region may comprise the processor **402**, **502** copying the low-activity region **304** to a reserved section (e.g., the reserved section **214** of FIG. 3) located in the block-addressable storage device **112** using a byte-based read command to read from the byte-addressable storage device **108** and a block-based write command to write to the block-addressable storage device **112** (block **620**). In some embodiments, the operations of block **620** for copying the low-activity region **304** may comprise converting from a byte format to a block format without losing data (block **622**). The processor **402**, **502** then updates a physical address of the low-activity region **304** in a translation table (e.g., the translation table **118** of FIG. 1) (block **624**).

(33) Next, the processor **402**, **502** copies the high-activity region **302** to a second reserved section (e.g., the reserved section **216** of FIG. 3) located in the byte-addressable storage device **108** using a block-based read command to read from the block-addressable storage device **112** and a byte-based write command to write to the byte-addressable storage device **108** (block **626**). Some

embodiments may provide that the operations of block **626** for copying the high-activity region **302** comprise converting from a block format to a byte format without losing data (block **628**). The processor **402**, **502** then updates a physical address of the high-activity region **302** in the translation table **118** (block **630**).

(34) It is to be understood that, in some embodiments, the operations of block **608** of FIG. **6A** for identifying the low-activity region **304**, the operations of block **612** of FIG. **6A** for identifying the high-activity region **302**, and the operations of block **618** of FIG. **6B** for exchanging the first storage region and the second storage region may be performed responsive to the processor **402**, **502** receiving the data access command at block **606** of FIG. **6A**. Some embodiments may provide that the operations of block **608** of FIG. **6A** for identifying the low-activity region **304**, the operations of block **612** of FIG. **6A** for identifying the high-activity region **302**, and the operations of block **618** of FIG. **6B** for exchanging the first storage region and the second storage region may be performed at periodic time intervals by the processor **402**, **502**. For instance, the processor **402**, **502** may execute a background process that wakes at regular time intervals and performs the operations of blocks **608** and **612** of FIG. **6A** and block **618** of FIG. **6B**. Some embodiments may provide that the low-activity region **304** and the high activity region **302** are identified by a host application processor (not shown) and provided to the processor **402**, **502** for execution without the use of the built-in statistics capabilities of the processor **402**, **502**.

(35) FIG. **7** is a block diagram of an exemplary processor-based device **700**, such as the processor-based device **400** of FIG. **4** or the processor-based device **500** of FIG. **5**, that migrates data between byte-addressable and block-addressable storage devices. The processor-based device **700** may be a circuit or circuits included in an electronic board card, such as a printed circuit board (PCB), a server, a personal computer, a desktop computer, a laptop computer, a personal digital assistant (PDA), a computing pad, a mobile device, or any other device, and may represent, for example, a server or a user's computer. In this example, the processor-based device **700** includes a processor **702**. The processor **702** comprises one or more general-purpose processing circuits, such as a microprocessor, central processing unit, or the like, and may correspond to the processor **402** of FIG. **4** or the processor **502** of FIG. **5**.

(36) The processor **702** is configured to execute instructions for performing the operations and steps discussed herein. In this example, the processor **702** includes an instruction cache **704** for temporary, fast access memory storage of instructions, and further includes an instruction processing circuit **710**. The instruction cache **704** stores instructions that are fetched from a memory, such as from a system memory **708** over a system bus **706**. The instruction processing circuit **710** is configured to retrieve the instructions fetched into the instruction cache **704**, and process the instructions for execution.

(37) The processor **702** and the system memory **708** are coupled to the system bus **706**, and can intercouple peripheral devices included in the processor-based device **700**. The processor **702** communicates with these other devices by exchanging address, control, and data information over the system bus **706**. For example, the processor **702** can communicate bus transaction requests to a memory controller **712** in the system memory **708** as a non-limiting example of a peripheral device. Although not illustrated in FIG. **7**, multiple system buses **706** may be provided, wherein each system bus constitutes a different fabric. In the example of FIG. **7**, the memory controller **712** is configured to provide memory access requests to a memory array **714** in the system memory **708**. The memory array **714** is comprised of an array of storage bit cells for storing data. The system memory **708** may be a read-only memory (ROM), flash memory, dynamic random access memory (DRAM), such as synchronous DRAM (SDRAM), etc., and a static memory (e.g., flash memory, static random access memory (SRAM), etc.), as non-limiting examples.

(38) Other devices may be connected to the system bus **706** in some embodiments. As illustrated in FIG. **7**, these devices may include the system memory **708**, one or more input device(s) **716**, one or more output device(s) **718**, a modem **724**, and one or more display controller(s) **720**, as examples.

The input device(s) **716** may include any type of input device, including but not limited to input keys, switches, voice processors, etc. The output device(s) **718** may include any type of output device, including but not limited to audio, video, other visual indicators, etc. The modem **724** may be any device configured to allow exchange of data to and from a network **726**. The network **726** may be any type of network, including but not limited to a wired or wireless network, a private or public network, a local area network (LAN), a wireless local area network (WLAN), a wide area network (WAN), a BLUETOOTH™ network, and the Internet. The modem **724** may be configured to support any type of communications protocol desired. The processor **702** may also be configured to access the display controller(s) **720** over the system bus **706** to control information sent to one or more display(s) **722**. The display(s) **722** may include any type of display, including, but not limited to, a cathode ray tube (CRT), a liquid crystal display (LCD), a plasma display, etc.

(39) The processor-based device **700** in FIG. 7 may include a set of computer-executable instructions **728** that may be executed by the processor **702** for any application desired according to the instructions. The computer-executable instructions **728** may be stored in the system memory **708**, processor **702**, and/or instruction cache **704** as examples of non-transitory computer-readable medium **730**. The computer-executable instructions **728** may also reside, completely or at least partially, within the system memory **708** and/or within the processor **702** during their execution. The computer-executable instructions **728** may further be transmitted or received over the network **726** via the modem **724**, such that the network **726** includes the computer-readable medium **730**.

(40) While the computer-readable medium **730** is shown in an exemplary embodiment to be a single medium, the term “computer-readable medium” as used herein may refer to a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of computer-executable instructions **728**. The term “computer-readable medium” shall also be taken to include any medium that is capable of storing, encoding, or carrying a set of computer-executable instructions for execution by a processing device and that cause the processing device to perform any one or more of the methodologies of the embodiments disclosed herein. The term “computer-readable medium” shall accordingly be taken to include, but not be limited to, solid-state memories, optical medium, and magnetic medium.

(41) The embodiments disclosed herein include various steps. The steps of the embodiments disclosed herein may be formed by hardware components or may be embodied in computer-executable instructions, which may be used to cause a general-purpose or special-purpose processor programmed with the computer-executable instructions to perform the steps. Alternatively, the steps may be performed by a combination of hardware and software process.

(42) The embodiments disclosed herein may be provided as a computer program product, or software process, that may include a machine-readable medium (or computer-readable medium) having stored thereon instructions, which may be used to program a computer system (or other electronic devices) to perform a process according to the embodiments disclosed herein. A machine-readable medium includes any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computer). For example, a machine-readable medium includes: a machine-readable storage medium (e.g., ROM, random access memory (“RAM”), a magnetic disk storage medium, an optical storage medium, flash memory devices, etc.), and the like.

(43) Unless specifically stated otherwise and as apparent from the previous discussion, it is appreciated that throughout the description, discussions utilizing terms such as “processing,” “computing,” “determining,” “displaying,” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data and memories represented as physical (electronic) quantities within the computer system's registers into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission, or display devices.

(44) The algorithms and displays presented herein are not inherently related to any particular computer or other apparatus. Various systems may be used with programs in accordance with the

teachings herein, or it may prove convenient to construct more specialized apparatuses to perform the required method steps. The required structure for a variety of these systems will appear from the description above. In addition, the embodiments described herein are not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of the embodiments as described herein.

(45) Those of skill in the art will further appreciate that the various illustrative logical blocks, modules, circuits, and algorithms described in connection with the embodiments disclosed herein may be implemented as electronic hardware, instructions stored in memory or in another computer-readable medium and executed by a processor or other processing device, or combinations of both. The components of the systems described herein may be employed in any circuit, hardware component, integrated circuit (IC), or IC chip, as examples. Memory disclosed herein may be any type and size of memory and may be configured to store any type of information desired. To clearly illustrate this interchangeability, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. How such functionality is implemented depends on the particular application, design choices, and/or design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present embodiments.

(46) The various illustrative logical blocks, modules, and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a processor, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), or other programmable logic device, a discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. Furthermore, a controller may be a processor. A processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

(47) The embodiments disclosed herein may be embodied in hardware and in instructions that are stored in hardware, and may reside, for example, in RAM, flash memory, ROM, Electrically Programmable ROM (EPROM), Electrically Erasable Programmable ROM (EEPROM), registers, a hard disk, a removable disk, a CD-ROM, or any other form of computer-readable medium known in the art. An exemplary storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. The ASIC may reside in a remote station. In the alternative, the processor and the storage medium may reside as discrete components in a remote station, base station, or server.

(48) It is also noted that the operational steps described in any of the exemplary embodiments herein are described to provide examples and discussion. The operations described may be performed in numerous different sequences other than the illustrated sequences. Furthermore, operations described in a single operational step may actually be performed in a number of different steps. Additionally, one or more operational steps discussed in the exemplary embodiments may be combined. Those of skill in the art will also understand that information and signals may be represented using any of a variety of technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips, that may be references throughout the above description, may be represented by voltages, currents, electromagnetic waves, magnetic fields, or particles, optical fields or particles, or any combination thereof.

(49) Unless otherwise expressly stated, it is in no way intended that any method set forth herein be construed as requiring that its steps be performed in a specific order. Accordingly, where a method

claim does not actually recite an order to be followed by its steps, or it is not otherwise specifically stated in the claims or descriptions that the steps are to be limited to a specific order, it is in no way intended that any particular order be inferred.

(50) It will be apparent to those skilled in the art that various modifications and variations can be made without departing from the spirit or scope of the invention. Since modifications, combinations, sub-combinations and variations of the disclosed embodiments incorporating the spirit and substance of the invention may occur to persons skilled in the art, the invention should be construed to include everything within the scope of the appended claims and their equivalents.

Claims

1. A processor-based device, comprising a virtualization circuit communicatively coupled to a byte-addressable storage device and a block-addressable storage device; the virtualization circuit configured to: present, to a processor of the processor-based device, the byte-addressable storage device and the block-addressable storage device as a single virtual storage device; generate a usage statistic for each of the byte-addressable storage device and the block-addressable storage device; identify a low-activity region in the byte-addressable storage device based on the usage statistic for the byte-addressable storage device; identify a high-activity region in the block-addressable storage device based on the usage statistic for the block-addressable storage device; and exchange a first storage region corresponding to the low-activity region and comprising a memory address region of the byte-addressable storage device with a second storage region corresponding to the high-activity region and comprising a block region of the block-addressable storage device; wherein: the single virtual storage device is configurably presented as one of a byte-addressable virtual storage device and a block-addressable virtual storage device; the high-activity region comprises a block-addressable storage region for which the corresponding usage statistic exceeds a value of a promotion threshold; and the low-activity region comprises a byte-addressable storage region for which the corresponding usage statistic is lower than the usage statistic for the high-activity region.
2. The processor-based device of claim 1, wherein: the virtualization circuit comprises a translation table; and the virtualization circuit is configured to exchange the first storage region with the second storage region by being configured to: copy the low-activity region to a first reserved section located in the block-addressable storage device using a byte-based read command to read from the byte-addressable storage device and a block-based write command to write to the block-addressable storage device; update a physical address of the low-activity region in the translation table; copy the high-activity region to a second reserved section located in the byte-addressable storage device using a block-based read command to read from the block-addressable storage device and a byte-based write command to write to the byte-addressable storage device; and update a physical address of the high-activity region in the translation table.
3. The processor-based device of claim 2, wherein: the virtualization circuit is configured to copy the low-activity region by being configured to convert from a byte format to a block format without losing data; and the virtualization circuit is configured to copy the high-activity region by being configured to convert from a block format to a byte format without losing data.
4. The processor-based device of claim 1, wherein: the virtualization circuit is further configured to receive, from the processor, a data access command to access the high-activity region; and the virtualization circuit is configured to identify the low-activity region, identify the high-activity region, and exchange the first storage region with the second storage region responsive to receiving the data access command.
5. The processor-based device of claim 4, wherein: the virtual storage device comprises a byte-addressable virtual device; and the data access command comprises a byte-based storage command.
6. The processor-based device of claim 4, wherein: the virtual storage device comprises a block-addressable virtual device; and the data access command comprises a block-based storage

command.

7. The processor-based device of claim 1, wherein the virtualization circuit is configured to identify the low-activity region, identify the high-activity region, and exchange the first storage region with the second storage region at periodic time intervals.

8. The processor-based device of claim 1, wherein the promotion threshold comprises one or more of an access frequency threshold, a data access latency threshold, an endurance threshold, and an access request size threshold.

9. A method, comprising: presenting, by a processor of a processor-based device, a byte-addressable storage device and a block-addressable storage device as a single virtual storage device; generating a usage statistic for each of the byte-addressable storage device and the block-addressable storage device; identifying, by the processor, a low-activity region in the byte-addressable storage device based on the usage statistic for the byte-addressable storage device; identifying, by the processor, a high-activity region in the block-addressable storage device based on the usage statistic for the block-addressable storage device; and exchanging, by the processor, a first storage region corresponding to the low-activity region and comprising a memory address region of the byte-addressable storage device with a second storage region corresponding to the high-activity region and comprising a block region of the block-addressable storage device; wherein: the single virtual storage device is configurably presented as one of a byte-addressable virtual storage device and a block-addressable virtual storage device; the high-activity region comprises a block-addressable storage region for which the corresponding usage statistic exceeds a value of a promotion threshold; and the low-activity region comprises a byte-addressable storage region for which the corresponding usage statistic is lower than the usage statistic for the high-activity region.

10. The method of claim 9, wherein exchanging the first storage region with the second storage region comprises: copying the low-activity region to a first reserved section located in the block-addressable storage device using a byte-based read command to read from the byte-addressable storage device and a block-based write command to write to the block-addressable storage device; updating a physical address of the low-activity region in a translation table; copying the high-activity region to a second reserved section located in the byte-addressable storage device using a block-based read command to read from the block-addressable storage device and a byte-based write command to write to the byte-addressable storage device; and updating a physical address of the high-activity region in the translation table.

11. The method of claim 10, wherein: copying the low-activity region comprises converting from a byte format to a block format without losing data; and copying the high-activity region comprises converting from a block format to a byte format without losing data.

12. The method of claim 9, further comprising receiving, by the processor, a data access command to access the high-activity region; wherein the method comprises identifying the low-activity region, identifying the high-activity region, and exchanging the first storage region with the second storage region responsive to receiving the data access command.

13. The method of claim 12, wherein: the virtual storage device comprises a byte-addressable virtual device; and the data access command comprises a byte-based storage command.

14. The method of claim 12, wherein: the virtual storage device comprises a block-addressable virtual device; and the data access command comprises a block-based storage command.

15. The method of claim 9, comprising identifying the low-activity region, identifying the high-activity region, and exchanging the first storage region with the second storage region at periodic time intervals.

16. The method of claim 9, wherein the promotion threshold comprises one or more of an access frequency threshold, a data access latency threshold, an endurance threshold, and an access request size threshold.

17. A non-transitory computer-readable medium, having stored thereon computer-executable

instructions that, when executed, cause a processor of a processor-based device to: present a byte-addressable storage device and a block-addressable storage device as a single virtual storage device; generate a usage statistic for each of the byte-addressable storage device and the block-addressable storage device; identify a low-activity region in the byte-addressable storage device based on the usage statistic for the byte-addressable storage device; identify a high-activity region in the block-addressable storage device based on the usage statistic for the block-addressable storage device; and exchange a first storage region corresponding to the low-activity region and comprising a memory address region of the byte-addressable storage device with a second storage region corresponding to the high-activity region and comprising a block region of the block-addressable storage device; wherein: the single virtual storage device is configurably presented as one of a byte-addressable virtual storage device and a block-addressable virtual storage device; the high-activity region comprises a block-addressable storage region for which the corresponding usage statistic exceeds a value of a promotion threshold; and the low-activity region comprises a byte-addressable storage region for which the corresponding usage statistic is lower than the usage statistic for the high-activity region.

18. The non-transitory computer-readable medium of claim 17, wherein the computer-executable instructions cause the processor to exchange the first storage region with the second storage region by causing the processor to: copy the low-activity region to a first reserved section located in the block-addressable storage device using a byte-based read command to read from the byte-addressable storage device and a block-based write command to write to the block-addressable storage device; update a physical address of the low-activity region in a translation table; copy the high-activity region to a second reserved section located in the byte-addressable storage device using a block-based read command to read from the block-addressable storage device and a byte-based write command to write to the byte-addressable storage device; and update a physical address of the high-activity region in the translation table.
